

COMM144 Generative AI 4th Workshop

Guoqiang Zhang

Exercise 1

Consider how to use the binary cross entropy loss (BCE) when updating the generator and discriminator.

Training procedure

Algorithm 2 Modified GAN Training

1: **for** number of training iterations **do**

2: **for** k steps **do**

3: Sample minibatch of m noise samples $\{z_1, \dots, z_m\}$ from $\mathcal{N}(z; \mathbf{0}, \mathbf{I})$

4: Sample minibatch of m real data samples $\{x_1, \dots, x_m\}$

5: Update the discriminator via SGD:

$$\theta \leftarrow \theta + \eta \nabla_{\theta} \frac{1}{m} \sum_{i=1}^m \left[\log D_{\theta}(x_i) + \log(1 - D_{\theta}(G_{\theta}(z_i))) \right]$$

6: **end for**

7: Sample minibatch of m noise samples $\{z_1, \dots, z_m\}$ from $\mathcal{N}(z; \mathbf{0}, \mathbf{I})$

8: Update the generator via SGD:

$$\phi \leftarrow \phi + \eta \nabla_{\phi} \frac{1}{m} \sum_{i=1}^m \left[\log(D_{\theta}(G_{\phi}(z_i))) \right]$$

9: **end for**

Exercise 2

Practice with python programming for training a GAN over MNIST in the python notebook.